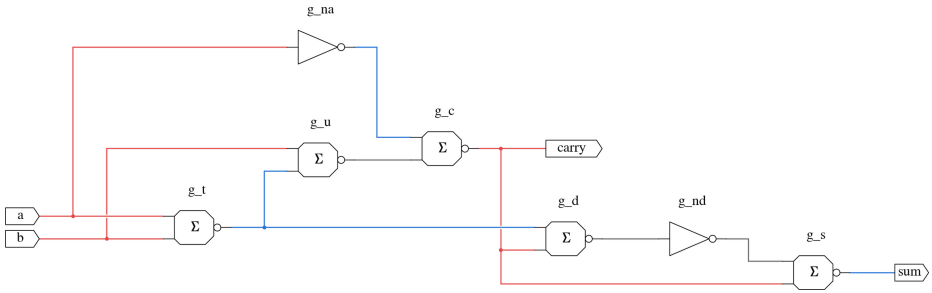
TritVerilog

```
full_add fa0(
  a[0],b[0],cin,
  sum[0],c1);
full_add fa1(
  a[1],b[1],c1,
  sum[1],c2);
full_add fa2(
  a[2],b[2],c2,
  sum[2],c3);
full_add fa3(
  a[3],b[3],c3,
  sum[3],cout);
```

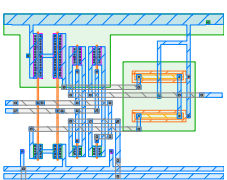
four full adders in a chain
+ full_add, half_add

Synthesis



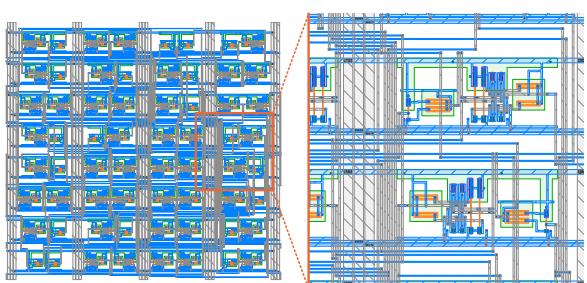
64 cells for the adder; one half adder at a = b = +1:
1 + 1 = sum −1, carry +1 (red +1, blue −1, gray 0)

Standard cell



NANY, from the taped-out
circuit; 3 rails

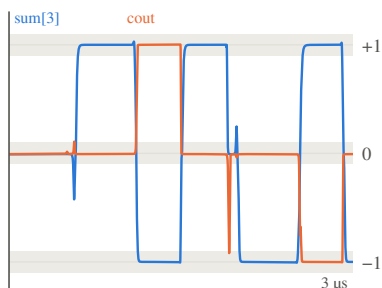
Place & route



1692 × 1590 µm, M1 + M2
DRC 0, LVS match

magnified: transistors,
M1, M2 and vias

Post-layout SPICE



extracted devices + wire C
three levels, ±0.5 V bands